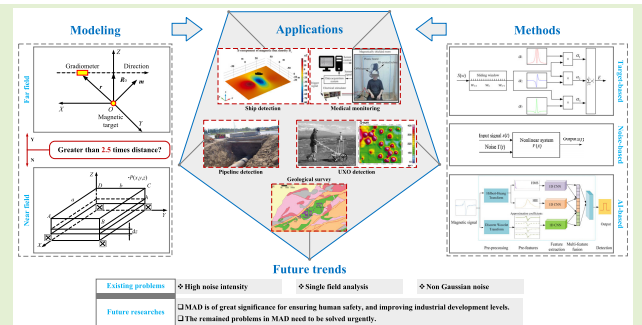


Theories, Applications, and Expectations for Magnetic Anomaly Detection Technology: A Review

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Abstract—Magnetic anomaly detection (MAD) is one of the most effective methods for engineering and environmental geophysical exploration, and plays an important role in scientific research and human technological innovation. Although numerous systematic and valuable articles have been published on the topic of MAD, they only briefly summarized relevant research from an algorithm perspective. Moreover, the practical applications in different scenarios are neglected and a comprehensive review of MAD is still lacking. In this case, it is particularly necessary to implement a comprehensive survey of the relevant publications of MAD, which could help readers to understand the current state-of-the-art techniques conveniently and quickly develop an effective solution to MAD problems in practice. Specifically, the developments of the MAD, mainly consisting of detection models and detection methods, are combed in this article. In addition, the different application scenarios of the MAD have been argued. More importantly, the existing issues and future trends are further discussed. We believe that the work done in this survey could supply convenience and inspiration for the researchers who want to devote their efforts to MAD.

Index Terms—Application scenarios, magnetic anomaly detection (MAD), magnetic dipole modeling, magnetic signal processing.



I. INTRODUCTION

THE spatial distribution of the original geomagnetic field will be changed by the ferromagnetic objects, forming the geomagnetic anomaly [1], [2]. Based on this phenomenon, magnetic anomaly detection (MAD) technology has been developed [3], [4], [5]. In addition, relevant information

about the magnetic target could be obtained by analyzing the magnetic signals measured by the magnetometer [6], [7]. Since the transmission of the magnetic field is unlikely to be affected by air, water, soil, and other media, the magnetic field signal is relatively stable [8]. Therefore, the MAD gets widespread use in geological survey [9], [10], [11], earthquake prediction [12], [13], [14], medical monitoring [15], [16], vehicle detection and positioning [17], [18], industrial nondestructive testing [19], [20], and so on, [21], [22], [23].

Generally, there are two critical points for the MAD including the model establishment and the magnetic anomaly signal detection. For the modeling, it can be divided into the far field and near field according to the distance between the magnetic target and the detector [24]. With the assumption that the distance between the target and the measurement point is greater than three times the size of the target, the spatial magnetic field distribution of the magnetic anomaly source is modeled as a single magnetic dipole [25], [26], [27]. Though the single magnetic dipole model has been applied more in real testing for its simplicity, there still remains a problem of limited detection accuracy [28], [29]. To solve such an issue, multipole model [30] and model function separation method [31] are proposed to forward model the

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magnetic targets with complex characteristics. However, when the condition of the aforementioned assumption is dissatisfied, regarding the near-field magnetic target as a magnetic dipole is improper. Hence, the theory of microphysics using the microelement molecular current, combined with numerical analysis and integral methods, is introduced to model the near-field magnetic source more accurately [32], [33].

The MAD methods could be summarized into three categories: target-based, noise-based, and artificial intelligence (AI)-based. The target-based MAD methods focus on analyzing the signature of the target signals, and its most typical method is based on the orthogonal basis functions (OBFs) [34], [35], [36]. However, the OBFs method is proven to retain well detection effect in the white Gaussian noise. As the magnetic environment gets complex, the performance of the traditional OBFs method is reduced significantly. Hence, amounts of the improved methods are proposed, including the OBFs based on prewhitening filter [37], the OBFs based on narrowband filter [38], and so on, [39]. In addition, the singular value decomposition (SVD) method [40], the principal component analysis method [41], and the time–frequency analysis method [42] are improved for MAD.

The noise characteristics are further analyzed to better extract the magnetic anomaly signal in the noise-based MAD methods. For instance, the minimum entropy detector (MED)-related methods utilize the energy entropy characteristic to suppress the noise [43]. However, the proper detection threshold is hard to determine. The energy detector based on empirical mode decomposition (EMD) trend filtering is developed [44]. In addition, there are other methods that consider the statistical characteristics of the noise, mainly based on the stochastic resonance (SR) [45], the high-order crossings (HOCs) [46], and the noise space analysis. To improve the SR performance in the condition of low signal-to-noise ratio (SNR) without using the target signal prior information, several improved SR methods are proposed by enhancing the target signal [47], [48]. To solve the problem of a small order of the traditional HOC method, an adaptive HOC detector was proposed [49]. Meanwhile, the adaptive coherent noise suppression (ACNS) [50] method and the noise space transformation (NST) [51] method are applied in real tests by analyzing the coherence and anisotropy of the noisy signal. Moreover, the MAD methods based on the sparsity and the low-rankness of the signal have also been reported, including sparse feature extraction [52] and low-rank decomposition [53], [54].

In recent years, with the progress of machine learning and deep learning technologies [55], [56], [57], [58], the MAD could be tackled as a classification issue. Hence, the multifeature fusion-based neural network [59], the support vector machine (SVM) [60], and so on, [61], [62] have been proposed. Nevertheless, these methods are still confronted with some unavoidable problems, including less feature information, complex detection network structure, and high requirements for training data.

To sum up, a great many researchers paid attention to the methods of noise suppression and target feature extraction. However, few kinds of literature comprehensively compare

and analyze the current MAD methods. Hence, the main contributions are outlined in the following four aspects.

- 1) Basic concepts and theories of MAD models are elaborated, including far-field modeling and near-field modeling, which presents a comprehensive overview of the MAD from the model perspective.
- 2) Three main categories of MAD including target-based MAD, noise-based MAD, and AI-based MAD are sorted out and analyzed systematically, and the corresponding basic concepts and theories are introduced, which presents a comprehensive overview of the MAD from the algorithm perspective.
- 3) Different applications of MAD are introduced from the perspective of practical scenarios, such as geological surveys, ferromagnetic target detection, and medical monitoring, which could instruct engineers to select the appropriate algorithms for specific applications.
- 4) A summarization of the current situations in MAD is presented, in which the existing problems and prospects of MAD are further discussed. Moreover, the future trends and potential expectations of MAD are concluded to provide new insights into the future works for seasoned researchers or potential newcomers.

Fig. 1 illustrates the overall logic of the literature review, the rest of this review is organized as follows. In Section II, the theoretical models of MAD, including far-field modeling and near-field modeling, are first briefly introduced to present how to build an effective forward model in MAD. Furthermore, three categories of MAD are elaborated and their basic concepts and theories are detailed from the perspective of the algorithms. Section III describes the recent developments in the field of MAD from the perspective of practical scenarios, in which five representative application scenarios are briefly introduced. More importantly, future trends and expectations on how to select appropriate algorithms for MAD in practical applications are shared in Section IV. Finally, the conclusions of this survey are given in Section V.

II. MODELS AND METHODS OF MAD

In this article, a framework is constructed for MAD as shown in Fig. 2, each part of which is elaborated in Sections II and III.

A. Theoretical MAD Models

1) *Far-Field Modeling*: For the MAD, it is necessary to establish a forward model for the magnetic target. Since the strength of the magnetic field decreases with the distance increases between the measuring point and the magnetic source. The typical single dipole, as shown in Fig. 3, could be constructed when the distance is $3\times$ larger than the size of the magnetic target [63], [64]. In this case, the magnetic field generated by the magnetic dipole is a function related to the distance and the target magnetic moment [65], [66], which can be expressed as

$$\mathbf{B} = \frac{\mu_0}{4\pi} \left(\frac{3(\mathbf{M} \cdot \mathbf{R})\mathbf{R}}{R^5} - \frac{\mathbf{M}}{R^3} \right) \quad (1)$$

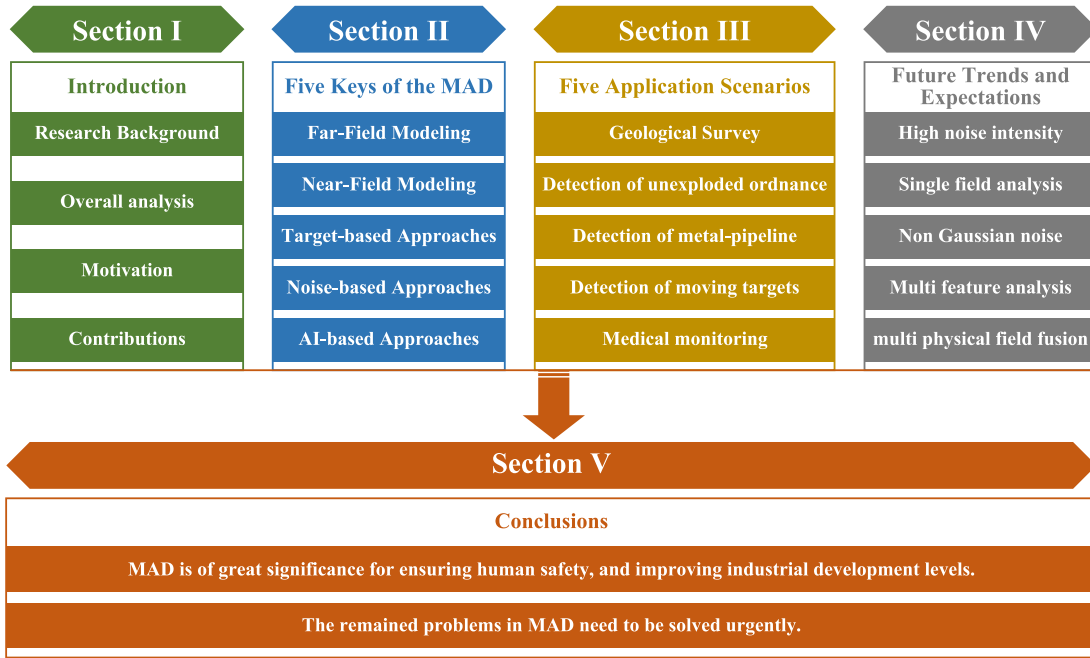


Fig. 1. Flowchart showing the overall logic of this literature review.

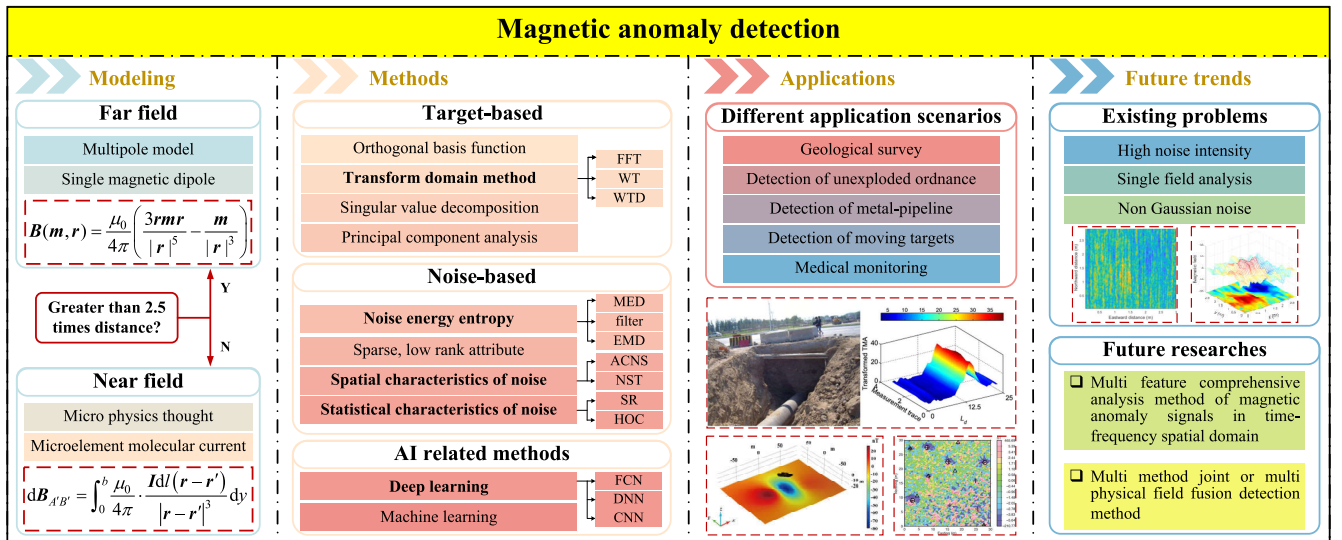


Fig. 2. Technical framework of MAD.

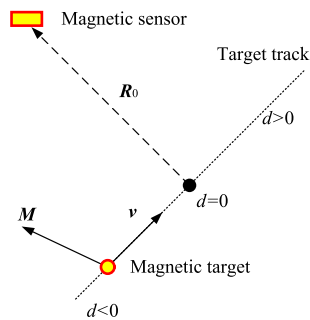


Fig. 3. Schematic of magnetic dipole model.

where μ_0 stands for the permeability of the vacuum, $\mathbf{M}$ stands for the magnetic dipole moment, and $\mathbf{R}$ and $R = |\mathbf{R}|$ stand for the position vector and the distance from the sensor to the magnetic dipole, respectively.

The measured magnetic field $\mathbf{B}_m$ is the vector superposition between the geomagnetic field $\mathbf{B}_e$ and the target magnetic field $\mathbf{B}_a$, and there is

$$\mathbf{B}_m = \mathbf{B}_e + \mathbf{B}_a \quad (2)$$

which can be rewritten in a scalar form as

$$B_m^2 = B_e^2 + B_a^2 + 2\mathbf{B}_e \cdot \mathbf{B}_a \quad (3)$$

where B_m , B_e , and B_a are the modulus of the corresponding variables. Generally, the magnetic field strength generated by the target buried underground is within tens of nT, which is far less than the geomagnetic field, i.e., $|B_a| \ll |B_e|$. In this case, (3) can be expressed as

$$B_m \approx B_e + \mathbf{e} \cdot \mathbf{B}_a \quad (4)$$

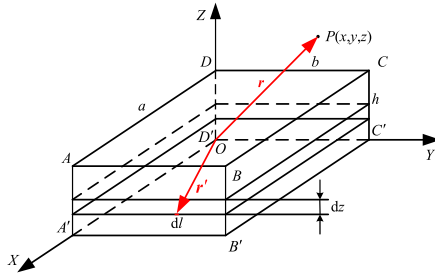


Fig. 4. Molecular current model of cuboid magnetized ferromagnetic objects.

where $\mathbf{e}$ is the unit vector of the geomagnetic field. Hence, the magnetic anomaly signal S can be expressed as

$$\begin{aligned} S &= B_{m1} - B_{m2} \approx (B_e + \mathbf{e} \cdot \mathbf{B}_{a1}) - (B_e + \mathbf{e} \cdot \mathbf{B}_{a2}) \\ &= \mathbf{e} \cdot (\mathbf{B}_{a1} - \mathbf{B}_{a2}) \end{aligned} \quad (5)$$

where B_{m1} and B_{m2} represent the magnetic field strength measured by the up and down sensors, respectively.

According to the single magnetic dipole model, a magnetic dipole location and tracking algorithm using magnetic gradient tensor data for direct inversion is first proposed [67], and then a numerical analysis model is established for magnetic anomaly target detection by assuming the submarine as a magnetic dipole and applied it to the detection of marine military magnetic targets [68], [69]. There are also some other location methods based on the magnetic dipole model, which requires a priori of some measurement parameters, e.g., magnetization angle [70], [71].

The single magnetic dipole is efficient to model the magnetic anomaly with a single magnetic source. However, when the shape of the magnetic target is complex, the real spatial distribution of the magnetic field could not be properly described. For instance, the error in calculating the magnetic field based on the magnetic dipole is discussed while the source of the errors is not considered [72]. Hence, a multipole model is constructed for a more precise description of the dual magnetic anomaly target, which expanded the range of detectable targets while improving detection performance [73].

2) Near-Field Modeling: For the near-field modeling, since the size and shape of the magnetic target cannot be ignored, to carry out more refined forward modeling of the magnetic target, the near-field magnetic anomaly model combined with the idea of microphysics is studied [74], [75], [76]. The normal near-field model is based on the theory of molecular circulation as shown in Fig. 4. Assuming the magnetization current density above the loop is $\mathbf{J}_s$, then the current intensity generated by the current loop $A'B'C'D'A'$ is $\mathbf{I} = \mathbf{J}_s dz$, and the $\mathbf{I}$ is a sheet current on the thickness dz . The origin of the space in Fig. 4 is the point O . Considering a measurement point P outside the cuboid magnetic target with a position vector $\mathbf{r}(x, y, z)$, taking a microelement dl of the edge of $A'B'$, and the magnetic induction intensity generated at point P is

$$d\mathbf{B}_{A'B'} = \int_0^b \frac{\mu_0}{4\pi} \cdot \frac{\mathbf{I} dl (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} dy. \quad (6)$$

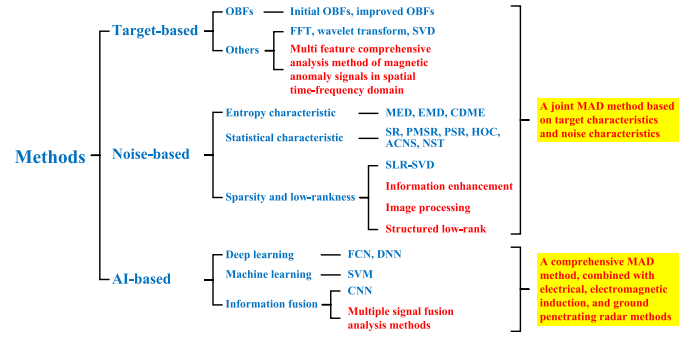


Fig. 5. Classification tree for the MAD methods.

Hence, the magnetic field generated by the total current loop $A'B'C'D'A'$ at point P can be written as

$$\mathbf{B} = \int_0^h d\mathbf{B}_{A'B'C'D'A'} dz. \quad (7)$$

For the MAD based on the near-field model, the underground pipelines were modeled based on the magnetic dipole and pipeline volume division by combining the far-field and near-field models [77], which could accurately reveal the intensity distribution characteristics of magnetic anomalies. In addition, the vector field distribution generated by a rectangular near-field ferromagnetic object has been simulated and verified, which provides an essential foundation for the MAD of the near-field target [78], [79].

B. Theoretical MAD Methods

In recent years, as the exploration depth deepened in MAD, the magnetic target and the environment are full of much more uncertainty and complexity, which may cause an extremely weak anomaly signal. In this case, the initial signal could be unavoidably drowned by various noises, such as current noise, electromagnetic interference, harmonic interference, and so on, [80], [81]. Hence, the SNR of the measured magnetic anomaly signal is extremely low, which brings a great challenge to enhance the detection performance. Therefore, a variety of MAD methods, roughly divided into target-based, noise-based, and AI-based, have been proposed successively. A tree to categorize the MAD methodologies is shown in Fig. 5, in which the mature applied methods are colored with blue and the methods to be further studied are colored with red.

1) Target-Based Approaches: The target-based approaches pay more attention to the features relevant to the magnetic targets, and then the anomaly signal could be obtained by analyzing the target energy features and the main components. For now, these approaches mainly consist of the OBFs method, the SVD method, and the principal component analysis method.

The classical target-based detection method is the OBFs and the corresponding detection model is shown in Fig. 6. Specifically, the movement of the sensor is along the D track, which is parallel to the X -axis. The dipole is located at the origin of the coordinate system. R_0 is the distance between the dipole and the track, and the $\mathbf{r}$ is the position vector

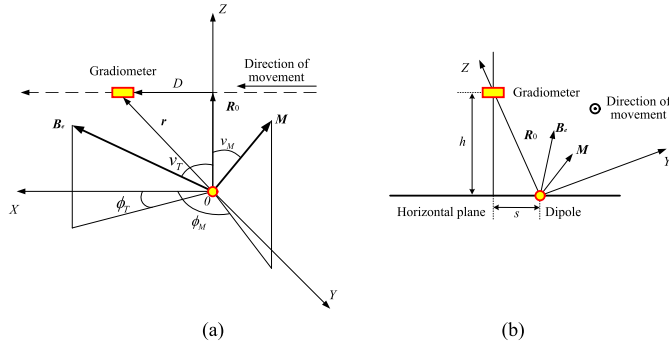


Fig. 6. Schematic of the OBFs detection model [34]. (a) View from the drawing plane. (b) View from the perpendicular plane.

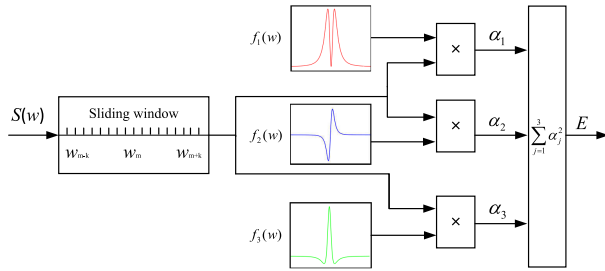


Fig. 7. Schematic of the OBFs method. The basis functions $f_1(w)$ – $f_3(w)$ are multiplied step-by-step by the raw signal flowing through the observation window. The three $\times$ blocks are the three multipliers. E is regarded as the energy of the raw signal $S(w)$ [34].

of the sensor. Fig. 6(a) is a view where track D lies in the drawing plane. B_e is the Earth's magnetic field. V_T and V_M are the angles between the Z -axis and the vectors B_e and M , respectively. ϕ_M and ϕ_T are the angles between the X -axis and the projections of the vectors M and T , respectively, onto the XY plane. Fig. 6(b) is a view where track D is perpendicular to the drawing plane. h and s , respectively, represent the height of the sensor and the horizontal distance of the dipole [34]. Likewise, the schematic of the OBFs principle is illustrated in Fig. 7. First, the proper OBF group is constructed according to Gram–Schmidt orthogonalization. Second, the original scalar-measured magnetic signal S_1 is projected into the OBF space to obtain the corresponding coefficients. Finally, the energy function is regarded as the decision function, and the magnetic anomaly signal could be detected by setting the appropriate energy threshold. The OBF space consisted of the typical three functions that can be expressed as

$$\begin{aligned} f_1(w) &= \sqrt{\frac{128}{3\pi}} \varphi_1(w) \\ f_2(w) &= \sqrt{\frac{128}{5\pi}} \varphi_2(w) \\ f_3(w) &= \sqrt{\frac{24}{5\pi}} \left[\varphi_3(w) - \frac{5}{3} \varphi_1(w) \right] \end{aligned} \quad (8)$$

where $\varphi_1(w) = (w^2 / ((1 + w^2)^{5/2}))$, $\varphi_2(w) = (w / ((1 + w^2)^{5/2}))$, $\varphi_3(w) = (1 / ((1 + w^2)^{5/2}))$, and $w = (D/R_0)$ is the dimensionless coordinate along the track of the magnetometer. Thus, the measured magnetic

signal S_1 can be expressed with the group of the OBF f_i and its coefficients a_i ($i = 1, 2, 3$) as

$$S_1 = \frac{\mu_0 M}{4\pi R_0^3} \sum_{i=1}^3 a_i f_i(w) \quad (9)$$

where R_0 is the scalar of R_0 .

Since the three-function orthonormal basis in (8) is more applicable to the scalar magnetic field signals, the feasibility of the five-function orthonormal basis to process the gradient magnetic field signals is further studied by Frumkis et al. [82], in which the number of the functions is simplified into four with more robustness. In addition, some filters were combined with the OBFs to improve the performance of the OBFs-based approaches in the non-Gaussian noise. For instance, the whitening filter based on the autoregressive (AR) model is introduced into the OBFs as a preprocessing procedure [83]. To tackle the deficiency of the poor detection ability in the $1/f^a$ noise, the OBFs detector based on the Karhunen–Loeve expansion is proposed [84]. Besides, the finite impulse response filter, which is able to eliminate high-frequency noise and preserve the information of the magnetic target signal, is combined with the traditional OBFs [85]. Moreover, to obtain the anomaly signal as clearly as possible, a full magnetic gradient detection method based on minimum entropy and gradient OBF is proposed [86]. However, it requires a relatively long processing time and prior information on magnetic targets.

In addition to the aforementioned time-domain MAD methods, the frequency-domain MAD methods were proposed by utilizing the frequency characteristic of the ferromagnetic target and the background noise signal lies in different frequency bands [87]. From this insight, the frequency-domain analysis methods, such as the wavelet analysis, the Fourier analysis, and so on, [88], [89], are verified in the MAD, and the SNR of the measured signal will be increased by maintaining the components of the spectrum coefficients. To improve the performance of the traditional OBFs under non-Gaussian noise, the wavelet decomposition method is first utilized to obtain the low-frequency component of the signal, and then the OBFs method is employed to further decompose [90]. Likewise, the wavelet denoising using an AR model [91] and the wavelet transform-based denoising methods are proposed to further suppress the power spectral density noise, which successfully eliminates the dependency of the noise signal variance by constructing a new threshold function that only needs to consider one parameter. Furthermore, to tackle the problem of the edge effect and external periodicity in standard fast Fourier transform (FFT) that limits the MAD accuracy, the nonuniform FFT (NUFFT) method is developed [42].

2) Noise-Based Approaches: In the noise-based MAD approaches, the magnetic anomaly signal is extracted by analyzing the characteristics of the noise, which mainly includes the entropy characteristics and the statistical characteristics.

The MAD methods based on the entropy characteristics typically consist of the MED and the entropy filter. According to the wavelet entropy theory, the improved wavelet threshold denoising method could effectively filter the noise contained

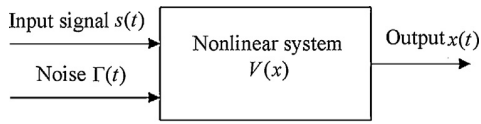


Fig. 8. Schematic of SR system.

in the magnetic anomaly signal by setting an appropriate entropy threshold [92]. Moreover, an improved MED detector using the information entropy is designed. It first estimates the magnetic noise probability density function of the differential signal by kernel smoothing, and then the anomaly could be obtained by the detection threshold. However, it is harsh to determine the threshold in different situations. Accordingly, the entropy filtering methods are verified in the MAD [93]. For instance, the anomaly can be detected according to the entropy filtering coefficients using the Neyman–Pearson criterion and the initial entropy value of the signal is calculated with a sliding window [94]. To further suppress the $1/f^a$ noise, an energy detector based on EMD trend filtering is applied in MAD [95]. This detector does not require any prior information about the target or background noise and performs well in detecting unknown ferromagnetic targets under unstable environments. In addition, an adaptive MAD method with ensemble EMD and minimum entropy features is raised aiming at the problem of the inconspicuous entropy features [26]. Furthermore, a coherent difference minimum-entropy (CDME) method is proposed to eliminate the common mode noise in magnetic anomaly signals, and then the SNR is increased by eliminating high-frequency noise through the hybrid differential minimum entropy method [96].

The MAD methods based on the statistical characteristics mainly consist of the SR [97], the HOC [98], [99], and the noise space analysis [100], [101]. Aiming to improve the MAD performance in the condition of low SNR and no target signal prior information, a new detector based on the SR [102] is proposed. Notably, from the point of utilizing the noise, the target weak signal is enhanced when the signal and the noise reach the matching relationship through the SR system, and the corresponding detection principle is shown in Fig. 8. The classical bistable SR system can be described by the nonlinear Langevin equation as $\dot{x} = ax - bx^3 + S + \Gamma$, where x is the output of the SR system, representing the displacement of Brownian particle in the adiabatic approximation theory [103]. S is the deterministic signal, Γ denotes the sum of external noise and inherent noise, and a and b are the system parameters characterizing the system. The symmetric double-well system potential is given by $V(x) = -ax^2/2 + bx^4/4$. When the bistable system is driven by simultaneous action of the signal and additive noise as shown in Fig. 8, it will be asymmetric. In addition, if the input deterministic signal, additive noise, and the nonlinear system cooperate well, the potential barrier may be reduced. At this time the particle that stays in one potential well, may surmount the energy barrier and enter another potential well, this phenomenon is called SR. When the SR phenomenon happens, the intensity of the useful signal is augmented, accordingly, the output of the SR system has a higher SNR in comparison with the input [104].

Nevertheless, the single SR system is vulnerable to the fluctuation of magnetic signals and sensitive to track selection. Hence, the parallel monostable SR (PMSR) and the parallel SR (PSR) are proposed to overcome the sensitivity problem and improve the detection stability [47], [48]. In addition, to tackle the challenge of real-time processing, the fourth-order aperiodic SR-based MAD method is further proposed [105].

HOC technology is a time series analysis method developed rapidly in recent years and was first applied in MAD by Sheinker et al. [106]. Specifically, first, this method analyzes the distribution characteristics of the magnetic background field to calculate the zero crossing number of the second-order difference sequence from the measured signal. Then, the anomaly signal could be separated from the noise according to the detection statistic of the zero crossing number. Thereafter, to tackle the problem of a low order of the HOC detection in the complex environment, an adaptive HOC detector is developed with the signal difference sequence extended to the fifth order [107]. In addition, the ACNS method is proposed to suppress the coherent geomagnetic background noise [108]. Compared with the synchronous reference subtraction [109], the ACNS could eliminate false target signals and further improve the capability of noise suppression. According to the spatial anisotropy of environmental noise, the NST method is proposed [110]. After the NST, the noise vector points will be distributed in a limited area, which can effectively suppress low-frequency noise.

In addition, with the development of sparse and low-rank theory [111], [112], the low-rank methods are gradually employed for signal processing, data analysis, computer vision, and so on, [113], [114], and have been applied to the MAD in recent years. For instance, to obtain the weak magnetic anomaly features submerged by the strong background noise, a sparse feature extraction method based on adjustable Q -factor wavelet transform and overlapping group contraction is proposed [115]. Compared with traditional wavelet transform, it is more flexible for adjusting the Q factor based on the oscillation characteristics of useful features. According to the compress sensing theory, an algorithm using the sparsity of magnetic dipoles in space is applied to locate magnetic targets [116]. In addition, a novel MAD method based on structured low-rank SVD (SLR-SVD) performs well in real scenarios by raising the low-rankness of the signal through the structural matrix lifting [117].

3) AI-Based Approaches: In recent years, with the development of machine learning and deep learning technologies, many researchers regard MAD as a binary classification problem [118]. In this case, the characteristics of the magnetic anomaly signals and the geomagnetic noise would be learned from a large number of training datasets. The anomaly could be detected through a cascade network, which consists of a binary classification detection network and a noise suppression regression network [119]. For instance, the MAD method based on the fully connected neural (FCN) network is proposed, in which the statistical characteristics of the signal and the magnetic moment characteristics of the target are taken as the input features, and the network structure with the optimal number of layers and nodes is finally obtained through iterative

training [120]. However, the performance of this method is worse under low SNR. Later, the deep neural network (DNN)-based MAD method is developed [121], in which the target signal can be automatically extracted and the noise irrelevant to the signal can be suppressed by learning the features from the collected data. However, due to the limited training dataset, the DNN method is susceptible to the type of the target and the noise intensity in practical applications, and the generalization ability is limited.

In addition to the neural network method, the SVM method is also widely used in the classification of magnetic anomaly signals. The MAD method based on the SVD and SVM, gets a 2-D real singular value matrix of the signal by the SVD, and thus the magnetic anomaly targets could be classified by the SVM [122]. What is more, the classification and recognition methods based on the SVM are combined with the wavelet packet denoising algorithm to further improve the detection accuracy [123]. Nevertheless, simply transforming the MAD into an issue of classification still remains the deficiencies of less useful feature information, complex detection network structure, and high requirements for training data. Besides, the performance might be seriously impacted when the power spectral density of the noise is $1/f^a$, where $0 < a < 2$ and f denotes the frequency.

In the past few years, with the development of information fusion technology, the multifeature fusion neural network is applied to the MAD to improve the detection effect of the FCN method under ultralow SNR [124]. Through comprehensively analyzing the features detected by three subclassifiers, including time–frequency features, statistical features, and magnetic moment features, the weighted voting method is then used for anomaly detection. Though the multifeature fusion neural network increases the detection probability compared to the FCN, the results are still sensitive to the direction of the magnetic target. Meanwhile, a MAD method based on multifeature fusion using a 1-D convolution neural network (CNN) is designed [122], and the framework is shown as Fig. 9. Specifically, the Hilbert marginal spectrum (HMS) and Hilbert instantaneous energy (HIE) as the feature of magnetic signals, which also contain physical meaning. In addition, to tackle the problem that the performance of 1-D CNN is easily affected by the low SNR signals, the strategy using 1-D CNN with multifeature fusion is designed to supervise learning on time-series signals to detect the weak magnetic anomaly and the detailed network architecture is shown in Fig. 10. It obtains the prefeatures of the magnetic signals in different dimensions through the Hilbert–Huang transform and the discrete wavelet transform, and then the magnetic anomaly signal could be recognized. In addition, to overcome the problem of high requirements of sample and poor practicability of supervised learning algorithm, a new MAD method based on feature fusion and the isolated forest is proposed [125]. This method not only requires fewer samples but has higher detection efficiency and shorter processing time.

III. DIFFERENT APPLICATION SCENARIOS

Up to now, the amount of researchers from academic or industrial communities has already brought extensive

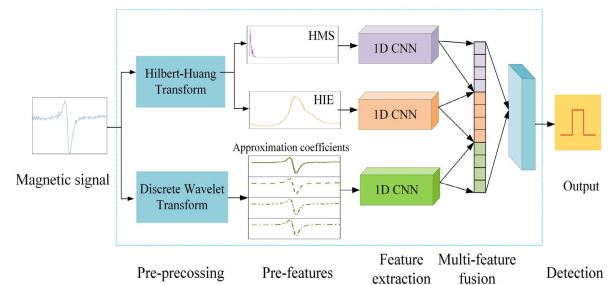


Fig. 9. Framework of the MAD method based on the convolutional neural network with multifeature fusion [122].

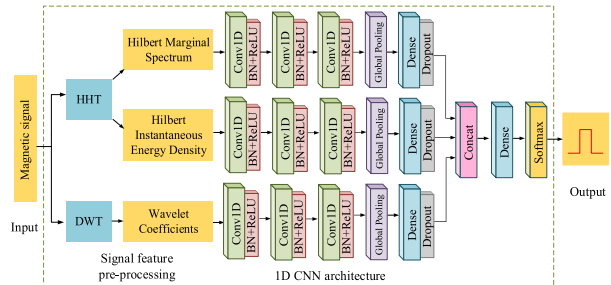


Fig. 10. Detailed network architecture of the convolutional neural network-based MAD method [122].

impressive achievements and successful field scenario applications using MAD technology. The MAD is initially employed mainly for detecting the military target in the Second World War, e.g., anti-submarine, and then is expanded to civilian gradually [126], [127], [128], [129]. With the improvement of the sensitivity, accuracy, installation flexibility, and environmental adaptability of the magnetic measurement equipment, the MAD is widely applied in geological surveys, earthquake prediction, medical monitoring, vehicle positioning, industrial nondestructive testing, and so on, [130], [131], [132], [133].

In terms of the geological surveys, the MAD is mainly used for resource prospecting and geological mapping [135], [136], [137]. For instance, during an investigation in Yushu, China, which is conducted by the Geophysical Exploration Institute of the General Administration of Metallurgical Geology of China [138], three new anomalies related to iron ore are found through the in-depth analysis of physical property data and interpretation of magnetic anomaly information. In addition, the geomagnetic anomaly was predicted from the measured EMAG2 data [139], and the distribution of different magnetic rock layers is investigated specifically in the geological survey in the Pobei area [134], whose test results are shown in Fig. 11. To reveal more details about underground structures, the full gradient tensor composed of nine potential tensor components is applied in this survey. Moreover, the corresponding normalization methods are further proposed to remove boundary noise and display more distinct geologic edges.

In terms of the unexploded ordnance (UXO) detection, the self-organized feature mapping (SOFM) [141], the magnetic anomaly structure index [142], and the automatic classification [143] are proposed and applied successively. For example, the UXO can be detected animatedly from broadband

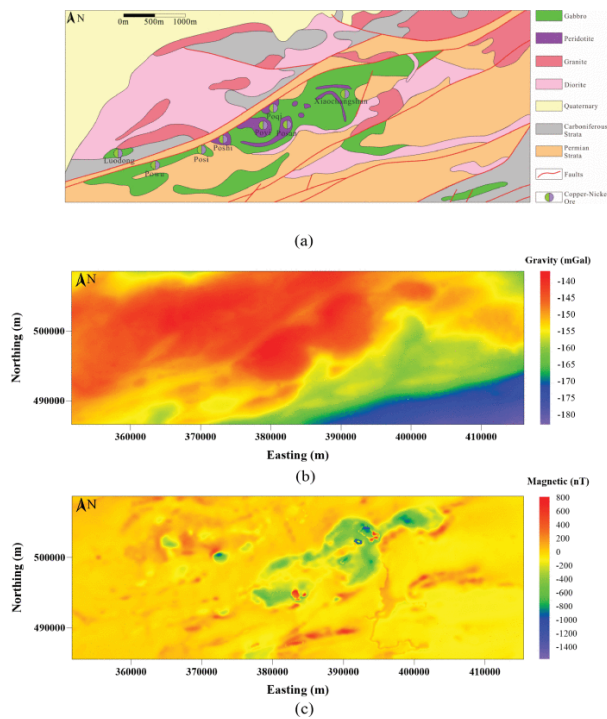


Fig. 11. Geological survey in the Pobei area. (a) Geologic map. (b) Bouguer gravity anomaly. (c) Reduced-to-pole magnetic anomaly of the study area with magnetization inclination of 60° and declination of 0.5° [134].

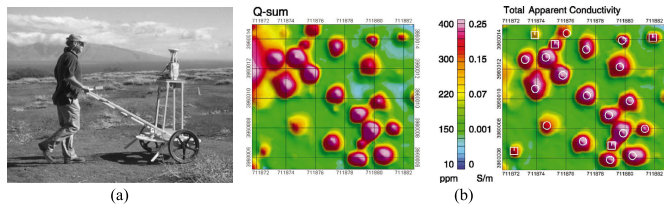


Fig. 12. UXO survey in Kaho'olawe, Hawaii. (a) Cart-mounted GEM-3 sensor. (b) Maps of Q_{sum} and TAC derived for the Geophex test site [140].

electromagnetic data [140]. To be specific, the data were collected at ten frequencies between 90 and 47970 Hz by a GEM-3 with a 96-cm diameter disk and a differential global positioning system mounted on a cart at 0.5-m line spacing. The sample rate is 30 Hz, yielding more than 20 data points/m at regular walking speed in a UXO survey, and the experiment condition and the target maps are shown in Fig. 12(a) and (b), respectively, where the Q_{sum} is the detector function that sums all quadrature data at all N frequencies. In addition, according to the extended Euler deconvolution, an automatic anomaly detection algorithm is put forward [144]. Among these methods, the self-organizing feature map neural network [141] and the discrete Hopfield neural network [145] are investigated to locate the UXO from airborne magnetic anomaly data. However, the issues of the massive training data and long processing time are still suffered temporarily.

In terms of the metal pipeline detection, the numerical integration [147], the finite element method (FEM) [148], and the magnetic dipole reconstruction (MDR) [149] are commonly used to build the forward model. To analyze the

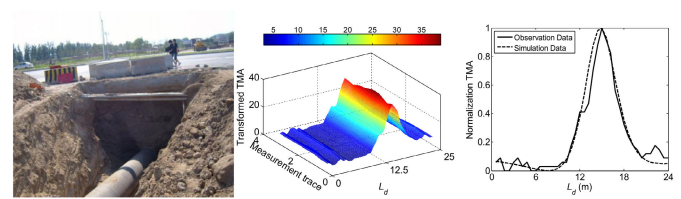


Fig. 13. Actual underground ferromagnetic pipeline and its total magnetic anomaly [146].

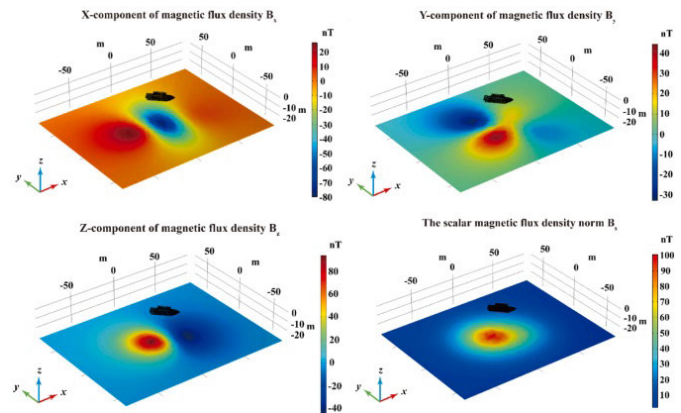


Fig. 14. Vector and the scalar magnetic flux density norm distribution induced by the movement of the ship [151].

influence of different factors on the detected magnetic anomalies, a numerical integration method is tested in Changping, Beijing, China [146]. It has performed a series of calculations for the model parameters, the geomagnetic field, and the measuring trace. The total magnetic anomaly with the corresponding distribution of an exposed pipeline is shown in Fig. 13. A low-cost wireless magnetic sensor positioning device is designed, and the performance of the MDR method is tested on an excavator, which demonstrates the superiority for estimating and locating the distance and depth of underground pipelines [150]. To improve the positioning accuracy, the magnetic sensor information is collected by the three-axis magnetometer to calculate the calibration coefficients of the linear relationship between the reference and measurement sensors. Therefore, the overall error rate for the distance estimation using the real-time pipeline detection system was about 25%.

In terms of the moving target detection, such as ships and submarines, the high precision modeling and the high operation speed of the MAD algorithms are essential. For instance, a 3-D FEM is applied for modeling the magnetic field distribution of the moving ships [151], with the vector magnetic flux density and the scalar magnetic flux density norm distribution induced by the movement of the ship as shown in Fig. 14. In this case, three packaged uniaxial magnetoelectric sensors are employed, whose ground noises are 6.04, 5.99, and 6.28 pT/Hz $^{1/2}$ at 1 Hz, respectively. The magnetoelectric sensors can detect magnetic fields as low as about 5×10^{-10} T, which represents a desirable performance for detecting extremely weak magnetic fields. In the case of real-time MAD approaches, an adaptive signal processing algorithm based on a superconducting quantum interference device (SQUID)

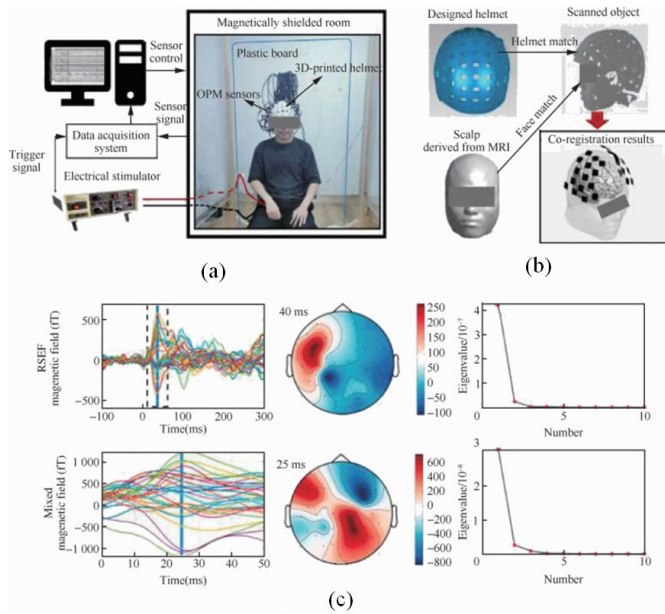


Fig. 15. Experiment process of the MEG. (a) Diagram of the OPM-MEG system as the acquisition equipment of the MEG. (b) Co-registration results of one subject. (c) Analyzing the MEG data through the somatosensory evoked fields, the topography of the component with the maximum amplitude, and the corresponding eigenvalues [157].

gradiometer system is designed, which can effectively eliminate motion noise [152]. Besides, to locate the magnetic targets fastly and precisely, the real-time positioning method of aviation total magnetic field based on the Eulerpsilas homogeneity equation is proposed and applied [153]. Whereas the detection of marine in the ocean environment, the EMD [154], the spatial signature analyzing [155], and the sliding window curve similarity (SWCS) [156] methods have also been verified.

In terms of medical monitoring, since there exist certain biological magnetic fields in the human body, including cardiac magnetic field, brain magnetic field, neural magnetic field, muscle magnetic field, and so on, the MAD for measuring the biological magnetic field is of critical importance in the human medical monitoring [15], [158], [159]. For instance, Jiancheng et al. [157] developed a MAD system based on optical-pumping magnetic sensors, and have obtained the magnetic field distribution of the brain and the heart using an extremely weak brain magnetic imaging device as shown in Fig. 15, which is effective for disease diagnosis related to the brain and heart. Specifically, the constructed 32-channel optical-pumping magnetometer (OPM)-based magnetoencephalography (MEG) system for MAD is used to measure cortical responses to median and ulnar nerve stimulations. To accurately map somatosensory cortex responses, a new method called free energy-based spatial smoothing estimation is proposed. The superiority of the proposed method is verified compared to the benchmark algorithm via simulations. In addition, this method is then applied to the source reconstruction of somatosensory evoked fields. The eigenvalues used for the neurons' source location are corresponding to the topography of the component with the maximum amplitude. The measured OPM-MEG data were analyzed at the sensor and source levels [15].

IV. FUTURE TRENDS AND EXPECTATIONS

Overall, Table I concludes the current solutions using the MAD approaches for the detection performance improvement from two models, i.e., far field and near field; from three methods, i.e., target-based, noise-based, and AI-based; and from five application scenarios, i.e., geological survey, UXO detection, metal pipeline detection, moving targets detection, and medical monitoring. Besides, the superiority and the weakness of the typical MAD methods are summarized.

From Table I, although extensive MAD methods have been proposed and applied in different fields, there are still some unsolved problems. For instance, most approaches only analyze the signal in a single domain. Because the characteristics of the magnetic anomaly signal in different domains cannot be analyzed simultaneously, there lies a specific limitation. Since the technologies of multiinformation fusion are gradually matured in recent years, a comprehensive multifeature analysis-based MAD through fusing the signal characteristics in the time, the frequency, and the spatial domains, will help to obtain more intrinsic properties. Likewise, it is necessary to combine with other methods, such as information enhancement, image processing, structured low-rank, and so on, to overcome the shortcomings and improve the efficiency and robustness of current MAD methods.

In addition, the target-based and noise-based MAD methods are all maintaining their own superiority and defect. The target-based methods ignored the impact of noise while focusing on analyzing the target characteristics, resulting in unsatisfactory performance under non-Gaussian noise. The noise-based methods are easy to cause noise suppression overfitting or signal distortion for the lack of information related to the real targets. Therefore, a joint MAD method based on target characteristics and noise characteristics is another essential research direction. Furthermore, with the development of magnetic field eliminations, nonmagnetic materials, and other industrial manufacturing technologies, the magnetic anomaly signals get weaker and weaker. Hence, a comprehensive MAD, combined with electrical, electromagnetic induction, and ground penetrating radar methods, is worth to be further studied. Likewise, the MAD accuracy could be further improved by thoroughly analyzing the magnetic, electrical, and electromagnetic characteristics of the corresponding signals.

Moreover, the magnetic sensors should be further developed for the effective detection of the ultraweak magnetic signal. From the aforementioned analysis of the magnetic sensors used in different application scenarios, the magnetic sensor with higher resolution, higher accuracy, and lower noise is urgently needed, and it could be further researched in the following aspects. The research of the new materials and preparation processes plays a critical role in the technology of high-sensitivity sensing. Hence, more attention should be paid to the strategies such as the novel single-spin quantum magnetic sensing based on diamond nitrogen-vacancy color centers, the spin-exchange relaxation-free atomic magnetic sensor, and the self-excited cesium optical-pumping magnetic sensor. More importantly, quantum magnetic sensors with new physical mechanisms and miniaturized magnetometers should be further investigated in the future.

TABLE I
KEY RESEARCH POINTS AND CLASSIFICATION OF MAD

Key points	Categorization	References	Common algorithms	Advantages	Disadvantages
Models	Far-field models	Oruç et al. [28], Pepe et al. [38], Wynn et al. [67], Tanekage et al. [68], Schneider et al. [70], Yang et al. [72]	Single magnetic dipole model, multi magnetic dipole model	Simple and easy to establish	Lower accuracy
	Near-field models	Yu et al. [32], Wan et al. [33], Guo et al. [77], Zhang [79], Li et al. [160]	Numerical analysis, integral	Higher accuracy and describe the target specifically	Complex operations
Methods	Target-based	Sheinker et al. [35], [36], [83], Zheng et al. [38], Frumkis et al. [82], Wan et al. [84], Yang et al. [85], Zhou et al. [86], Nie et al. [87], Zhang et al. [90], Zhou et al. [91], Wang et al. [42], Nie et al. [161]	OBFs, FFT, wavelet transform, SVD, PCA	Noise suppression in the frequency domain and well performance in the white Gaussian noise	Vulnerable to the colored noise and hard to determine the threshold
	Noise-based	Tan et al. [92], Sheinker et al. [94], Han et al. [95], Fan et al. [26], Wang et al. [96], Tang et al. [162], Liu et al. [50], Bayram et al. [97], Liu et al. [102], Liu et al. [47], Wan et al. [48], Sheinker et al. [106], Zhou et al. [107], Shen et al. [110], Hu et al. [115], Gijssels et al. [116], Zhang et al. [117]	MED, EMD, CDME, SR, PMSR, PSR, HOC, ACNS, NST, SLR-SVD	Filter the noise effectively and further enhance the target weak signal strength	Time-consuming, vulnerable to magnetic signal fluctuation, and sensitive to track selection
	AI-based	Zheng et al. [40], Xu et al. [119], Liu et al. [120], Wang et al. [121], Wang et al. [123], Xu et al. [124], Fan et al. [122], Zhang et al. [125]	FCN, DNN, SVM, CNN	Extract the target signal and suppress the noise automatically	Limited training dataset and generalization ability, and poor practicability of supervised learning algorithm
Applications	Geological survey	Mark et al. [135], Ferreira et al. [136], Wang et al. [138], Irma et al. [139], Li et al. [134]	Aeromagnetic survey, Earth magnetic anomaly grid, second vertical derivative	Analysis of the geomagnetic anomalies efficiently	Hard to be applied in complex geographical environment
	Unexploded ordnance	Tobely et al. [141], Nazlibilek et al. [143], Huang et al. [140], Reid et al. [144], Salem et al. [145]	SOFM, Hopfield neural network	Classify the anomaly automatically	Massive training data and long processing time
	Metal pipeline	Zhao et al. [147], Pan et al. [149], Guo et al. [146], Kim et al. [150], Li et al. [163], Guo et al. [164], Park et al. [165]	Numerical integration, FEM, MDRM, MDR	Low-cost and wireless sensing	Unable to describe the target in a deep burial depth
	Moving targets	Chen et al. [151], Humphrey et al. [152], Xiao et al. [153], Hasan et al. [154], Zhou et al. [155], Wang et al. [156]	Euler's homogeneous equation, EMD, SWCS	Locate the magnetic targets in three dimensions	High requirements for the modeling accuracy and the operation speed
	Medical monitoring	Zimmerman et al. [158], Cao et al. [159], An et al. [15], Yang et al. [166]	Variational free energy based spatial smoothing estimation, independent component analysis, EMD	Obtain more internal organ information	High requirements for the measurement environment and time-consuming

V. CONCLUSION

MAD is of great significance for promoting geoscience research, ensuring human safety, and improving industrial development levels. In view of the above synthetic investigation, the existing MAD models for the far field and near field are simply aimed at the pure magnetic field generated by the target, ignoring the geomagnetic and other background fields. Hence, constructing a MAD model closer to

the real-world magnetic measurement background is desirable. Moreover, the poor performance of the target-based MAD methods in non-Gaussian noise and the over-processing or signal distortion of the noise-based MAD methods, indicate that there is no one-size-fits-all solution for magnetic anomaly signals or different application scenarios. Furthermore, the recent advances in AI may tremendously promote MAD performance.

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